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Since we know ki SAP Business Solutions ko provide
Karta hai, & Unhone Sabko Cloud pe Shift Kar
Ka Plan kiya.

SAP BTP based hai AWS GCP Architecture pe jo ki Various Solutions provide bhi
Karta hai Aur hum design bhi Kar Sakte hai
Apne solutions.

Basically SAP with partnership in different
companies like Amazon AWS GCP Architecture
they built BTP & open for everyone in
2021

BTP uses Cloud foundry based on AWS GCP
Architecture → Hyperscaler's Infrastructure
↓ ??

Hyperscaler's - So many Scaling is possible.
U Imagine, We got it.

BTP → PaaS in which You can
create your Application as well as
also use some pre developed solutions.

Technical Sol vs Business Soln

↓
Developers
Cyber Sec
Technical Consultant
Devops

↓
Payroll soln
Business soln
HR soln

→ offered by BTP

BTP Cockpit

Entitlements

→ Service Assignments

You Can View all the Services

Business Soln - SAP Cloud Identity Access Governance

↳ If a user want access to any system
He can generate a request, once
approved automatically new acc
will be created.

QCP Cockpit

Global Account - Ek tarike ki Legal Contract firm
ho ki kaun kaun si services chahiye.
kitne amt mein chahiye kya pricing hai vo
sab discuss hone ke baad SAP will create an Global
Account.

SAP will add all the services to Global
Account.

Entitlements > Service Assignments

To run a service we need Infra Structure. So
we can run it.

Create a Subaccount

Connection b/w service & Infra so I can figure where I need to run this service US, Brazil, ----- & so-on

To Target a Location

Where I want to run Service.

You create a Subaccount such that it is closer to user

All the services are in Global Account You need to transfer it to Respective Sub Account

Entitlements > Entity Assignments.

You can see some default & free services.

Some paid Services need to be added

//_

Add Services plan → Add

Go In To Subacc

Here we can see a Services Section
You can see all the services as in Entitlement Section.

You can see your Instances & Subscription Access.
You need to go in services section for subscribe to Subaccount.

As soon as you subscribe any service or Application the roles are available

Security > users > Role collections - [...]

Build Work Zone → There are Roles like
Launchpad - Advance - Training
Launchpad

Global Acc

- Sub Acc — diff location
- Sub Acc — Australia
- Sub Acc — India
- Sub Acc — Singapore.

BZP Cockpit

is a logical Representation of the entire Landscape.

What is running where.

BZP is a Centralised Technical platform in which we have lot of technical service

Resource Provider (Database for AWS & AZURE)

If You have some resource already on AZURE & AWS You can use that

Database . You need to connect it.

This service is also provided.

//_

First You need to add that Resource let
suppose You are using Postgres SQL.

Then to Consume it You must have a
Service attached to a subaccount. So You
need postgres SQL.

Success factor - service] — same sub acc
postgres SQL - service]

Boosters - Step by Step Guided process.

I AAC - Infrastructure as a code.

Yamini code file in a way jiski up
Infra generate karo generally Yaml files hote
hai.

Bas

Similarly SAP have some script written so
we can create some

Creation of ABAP Env.

You can only assign Role Collection

Accordingly You can create Role Also

Imp points:

When You use BWP u need ext Identity provider with openID connect

I use cloud Identity Service for the same

Trust Yaani Apki Authentication ke Liye
SAP kisko contact karega.

You need to add a trust.

//_

AT a Global Acc Level

We can Create our Custom Role Collection.

We Cannot Create Role. We only can Create Collection.

We can only assign Role Collection to a user not Role

Trust Config

If You have Your own custom user directory we can provide it.

Custom Identity provider → we can use.

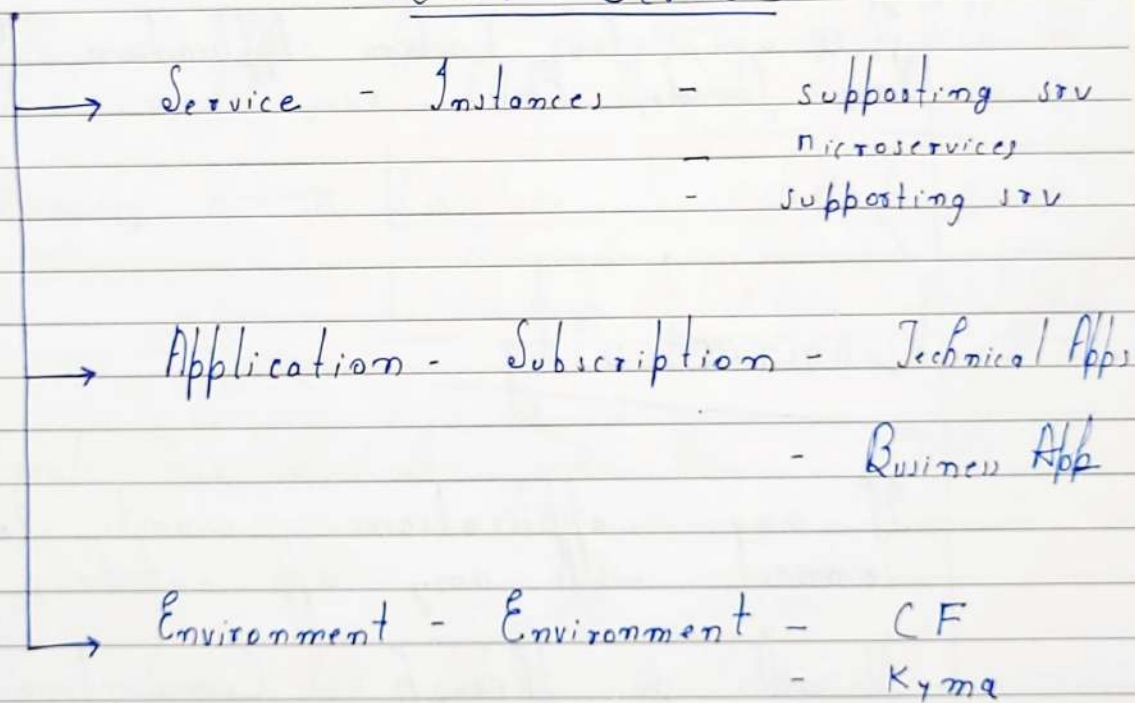
Sub App Level

Tenant - ?? $\Rightarrow$ Suppose there are 50 companies all are procuring Ariba. Now Ariba will have different instances for each company. It provide different tenant.

this is called Tenant.

Ariba $\rightarrow$ multi Tenant Application.

RCP Services



If we want to develop a custom app we will use cloud foundry as an Env

But if i am using an application then it is running on shared cloud foundry which is owned by SAP itself so make sure if there is some

Instance - support srv → should not use Your foundry for the application should use SAP cloud foundry

It is only for Custom Application should use Your foundry Env.

Connectivity

If any application wants to connect with any app outside Acc

It will go through Connective Nav

Directories.

For each service You will create sub
- acc

BW2 -

ABAP Env -

Hana Cloud -

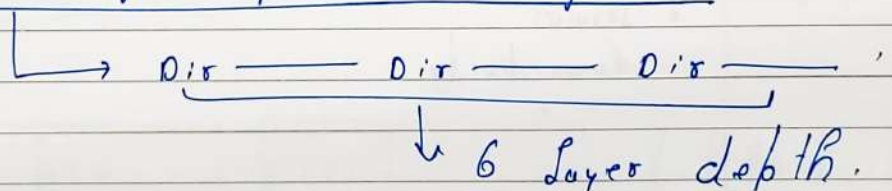
If there are 100+ Global Account So difficult to manage.

They are like Global Account which help You to model your bTP Account Effectively.



Create Your Landscape like that.

You Can Go Up to 6 Layers.

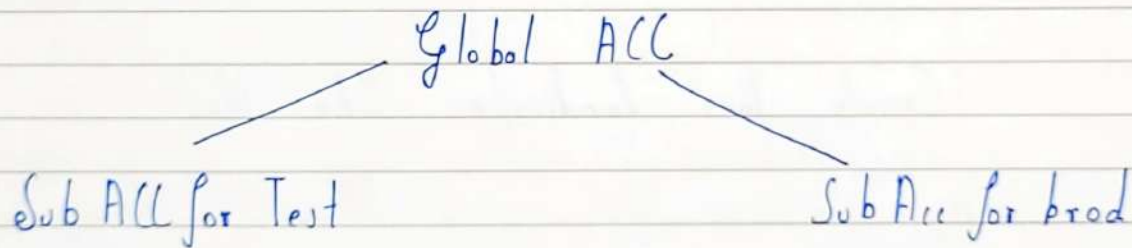


By default Entitlement & User Management is not active we need to activate

Global Account assign Entitlement to Directory then Directory Admin manage after that

Acc Model

1 Suitable for SAP Pre Delivered Business Solution Implementation



- Apps
- Services
- Connectivity
- "

"

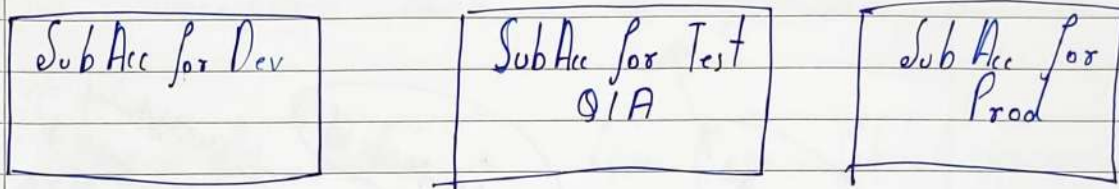
"

"

"

→ Functional Configuration is possible No custom development then 2 Tier Architect is fine.

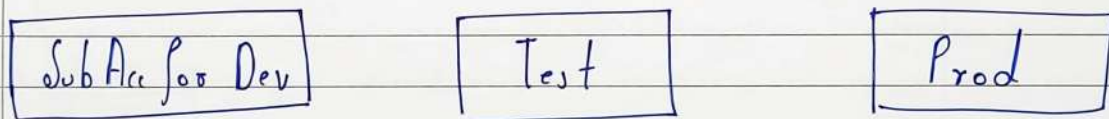
2 Custom Application Development



3 Data & API Seprate — they are part of backend

code

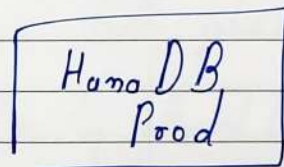
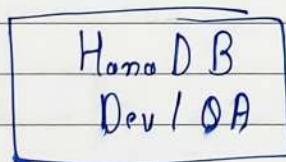
D1



Data

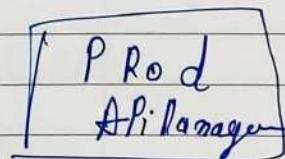
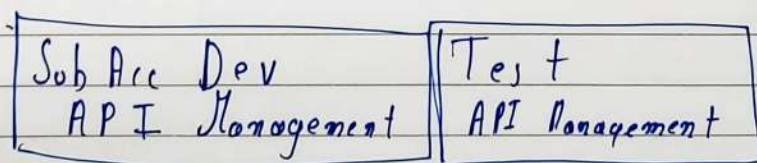
D2

Data



API.

D3



BCP Commercial Model

- > Consumption based - pay as you go, CPEA
- > Subscription based
- # If i am absolutely sure what I want go for subscription based. No Negotiation possible.
- # Commercial Model - Apply on Global Account.
- # Financial Model of a Global Account Can be changed. If and only SAP directly comes in picture. You need to discuss them. How & What will be perks. SAP will come.

Security

Authentication

Tells who

SAP IAS / Idp

Authorization

Are You allowed to do tasks

Tells what
XSUAA

Authentication

> Identity provider - Basically user ki Identity ko verify karke hai.

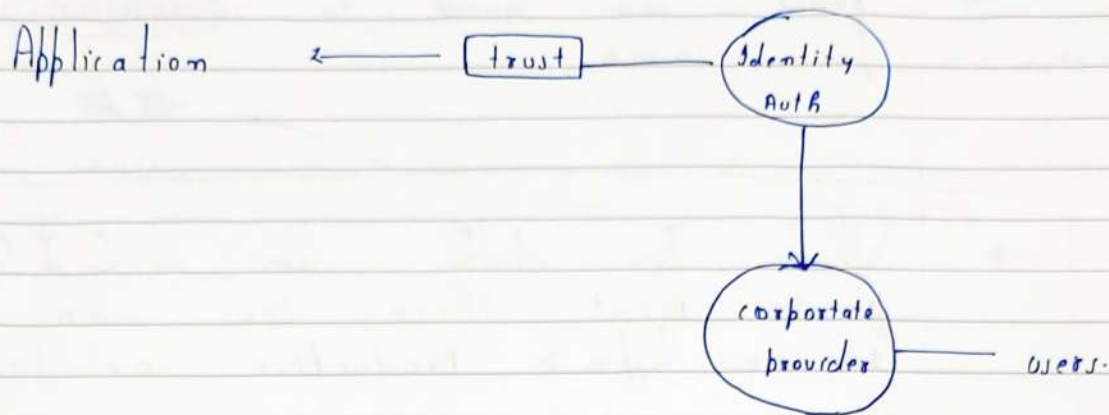
- Knowledge - username & password
- Possession - smartphone, You have
- Qualities - fingerprint or Retina Scan

May use any one of above to verify.
Can also use MFA - Multiple - factor Authentication.

You know abt Cloud Identity Services.

> If you want to check all tenants in your landscape.

= iam tenants - accounts - cloud.sap



Earlier before BTP people were using Azure, Cloud Identity provider SAP said keep this in between.

Cloud Identity Services.

- > IPS
 - > IDS
 - > IAS
- } Three Component.

→ You know You can create CIS Account.

//_

→ Now you need to establish trust for same

→ ★★

* When you click on CIS go to next there you can choose service type > Productive or test

Why we need Test & Prod IAS.

→ All dev & test system will trust Test IAS.

→ All prod system will trust Prod IAS.

IAS Trust

- > SAML Protocol
- > Open Id Protocol.

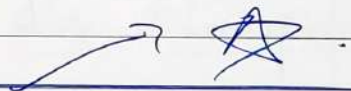
SAML

Basically in this system configuration files are shared Among Identity provider & Service provider. Meta data

Open-Id - based on OAuth

Request $\longrightarrow$ SP $\longrightarrow$ redirect $\longrightarrow$
IDP $\longrightarrow$ Login page

$\longrightarrow$ JWT $\longrightarrow$ Token
Check Token is generated for user & will get back to SP. , Token has a lifetime.



#

For open Id Connect You need diff URL's for configuration

Applications & Resources $\longrightarrow$ openId Connect Configuration Tenant settings $\longrightarrow$ Single Sign On

How You will Configure an App to IAS.

→ SubApp → Trust Configuration

Parameter -

Origin Key - Each Trust Configuration has a technical Identifier called origin key.

If there is a user in IAS but not in subapp he log in so automatically his profile will be created. then we can assign some role to it.

→ we can delete it in RCP but It will create again.

→ How this happen IDP sent this info in a packet to Application like

first, last name, emailId, others.

//_

This details that IAS sent we can send it.
control it.

Application & Resources.

Application (Bundled) > Trust > Single Sign On

↳ select that application

Single Sign On → Attributes

You can customize which things you want to send.

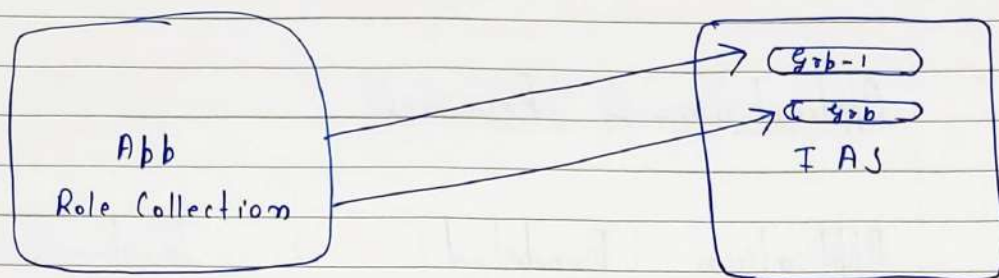
Automation of Role Assignment

★ We can also disable the setting of not creating shadow users.

Configure Tenant in Trust Configuration.

↳ Configure param. ☒ Bottom a check box

In IAS we can create Group.



Role Collection-1 → Grp-1
 - 2 Grp-2

We Can Map this

IAS

Users & Authorization. ✓ → create Group

★ Restrict (Optional) for a particular Application.

★ Add user in a Grp.

BCP

Trust Configuration > click trust url > Role collection mappings

Role collection — which role

Attribute — Group

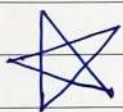
op₂ — equals

value — Name of Grp NOT Display name.

Now Application will receive sname, email, but it also needs Grpname

SAML

Configuration.



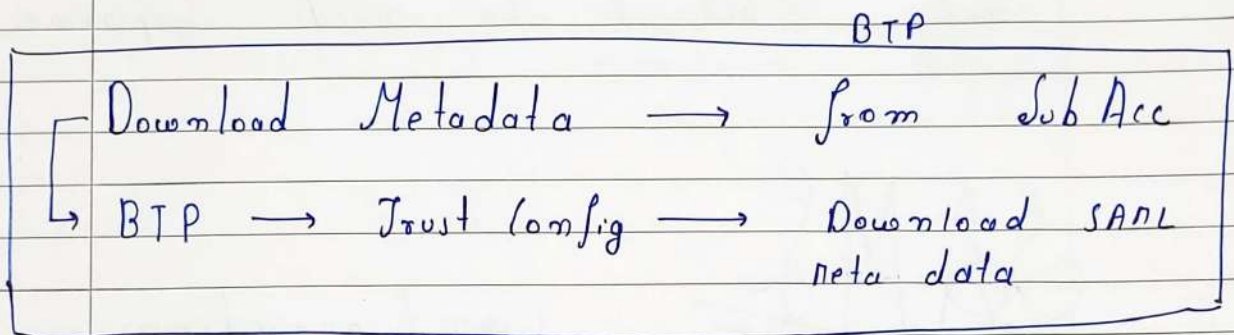
You Cannot have same trust with OpenId & Saml. means same IDP cannot have both protocols.

You know for this meta data needs to be shared.

IAS

Application & Resources > create

Homeurl → Get from Instance & Subscription. → url of application.



SAML 2.0 Configuration ↓

Define from meta data

IAS Knows Application but

Now Vice Versa.

//_

Application & Resources → Tenant Settings
→ single sign on → SAML 2.0 configuration

> Download Meta data file

BTP

New SAML Trust Configuration > upload data

Link Text => when login page comes it shows option

#

Important Note => Once we created app with SAML. by default Group data will not be sent.

In Grp don't get Confuse Grp Name & Display Name

→ Display Name

select ← Grp Name

??

Imp

I Cannot see a Role Collection in user for shadow user but then also I am able to access

Cause User does not have Role

Grp has Role

User $\longrightarrow$ Group $\longrightarrow$ Role Collection

Federated Authorizations.

Subject name - Identifier - This will uniquely identify a user

Fall back - Identifier - Alternate Attribute if Primary attribute is Null.

★ IAS use Cases

- As Main Identity provider
- Proxy Identity provider

In IAS BCP Cockpit option to assign role is

Trust Configuration > or Attribute Mappings

Security > Role Collection. → ★ ★

Global ACC

Trust Configuration > Only openId possible

As soon as we create trust at Global Acc level

It is copied in every sub Acc & Directories.

> Custom Identity provider for platform users

→ It will come under this heading

→ This trust will provide access to subacc

Now what link you provide of SubAcc for Global Acc

Goto > Trust Configuration > platform users >
SAP BTP Cockpit > open

Authorization

Same You to Go SubAcc > Role
Collection > Platform Roles > Add in
user groups.

SAP Recommends openId to use They are depreciating SAML.

Migration

If we have a trust in SAML want to go to open Id.

We Can do this through CLI Not UI

What If

By mistake someone delete a trust.

Get timestamp from Audit Logs. Then Contact SAP for backup

ENVIRONMENT

When I say Environment where my app
can run

RunTime Env - Java, ABAP, Ruby

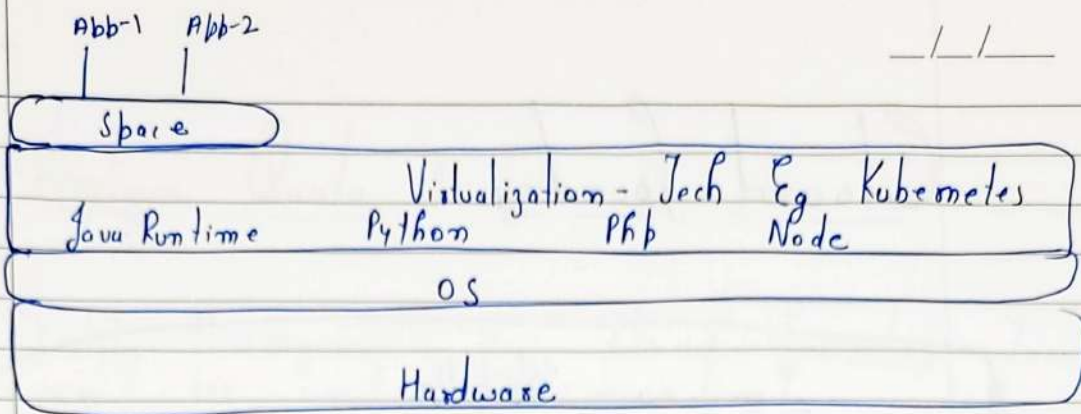
OS

CPU

RAM

→ All this together

BZP



→ Smart.

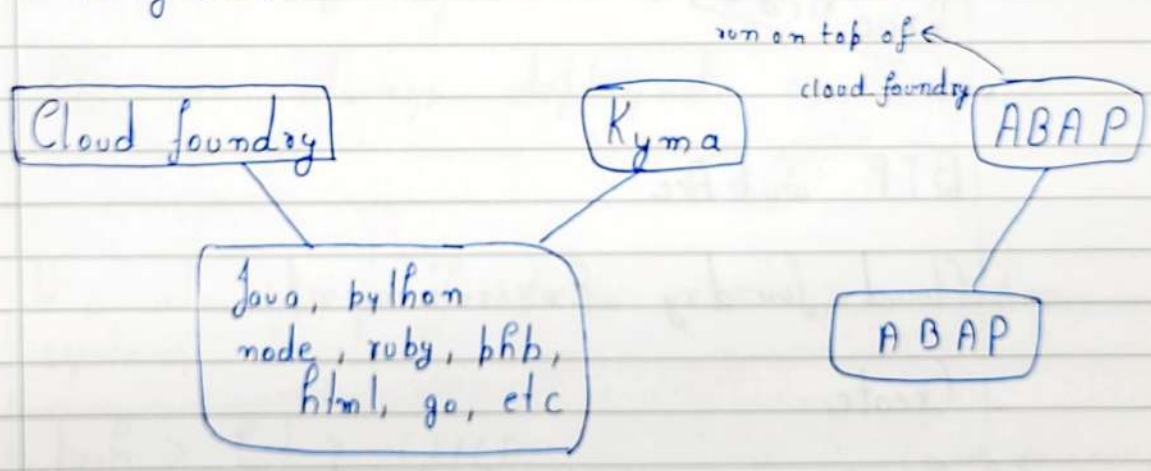
Kubernetes - Automatically checks manifest file & download Necessary things & Run that Environment

Available

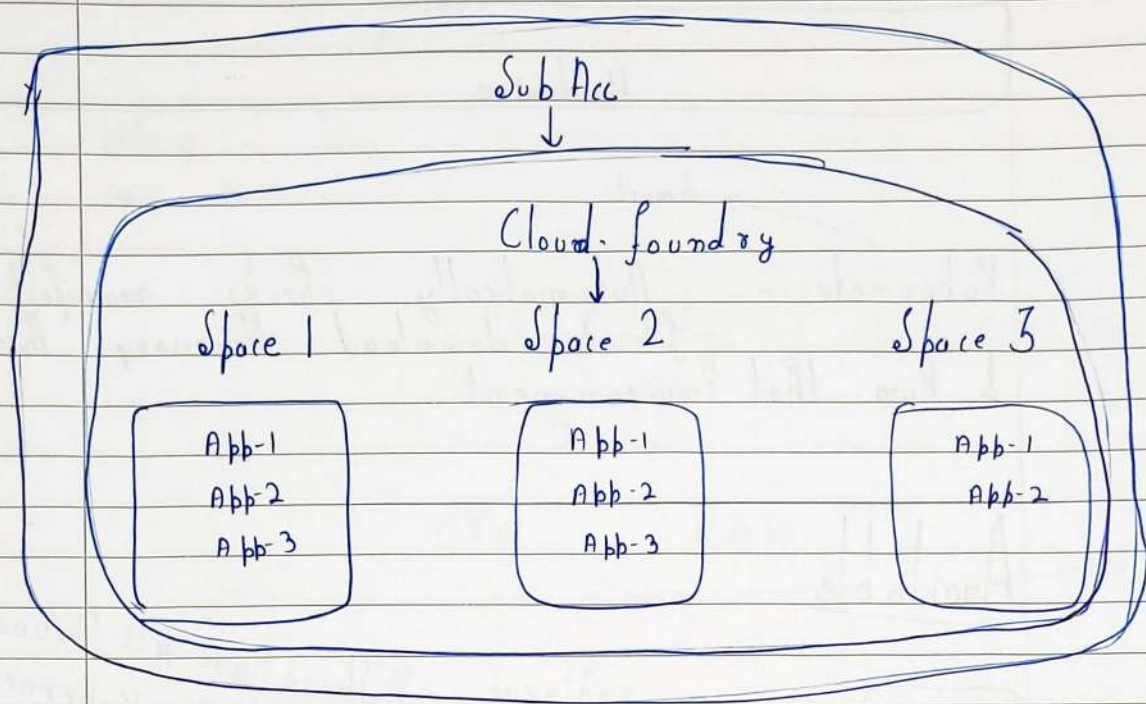
NeO

Retiring Soon

These are all flavour of Kubernetes
Powered by ↓



Cloud foundry



Practical

BTP Sub Acc

Cloud foundry Environment.

Create.

| Sub Acc > | Cloud foundry Env.

Assign Quota.

Entity Assignment > cloud foundry You
can see unit. Add accordingly

Add Space

SubArc > General > Create Space

Navigation - Cloud foundry > Spaces

- # You Can Logically Create partition.
- # Apps are always deployed on Space
- # It is always preferred to use a proper system
bash > Git > CI/CD > CTMS > Code deliver

By default it will reserve 1 GB

Override:

cf push <app-name> -m 256 M -k 256 M

memory disk

small 量

You can also limit a space like
how much or Using Quota

Space Quota - Create

Routes means ?

/ idx.html	} all Endpoints
/ page1.html	
/ page2.html	

Services - How many services You can bind.

//_

If we find load increased we can create instance of app.

We can also automate it.

Instance Memory is this.

Set all parameters > Create a Spare Quota.

Allow paid Service > on = If business critical app
↳ pay more but business doesn't affect

Quota Assignment

★ AT a Spare. we can also see that services tab here we have all services that require an app.

Instance

We subscribe to connectivity service.

We need to create binding to app.

SubAc Level > Instances & Subscription >

Click there side panel bound Application
create.

or

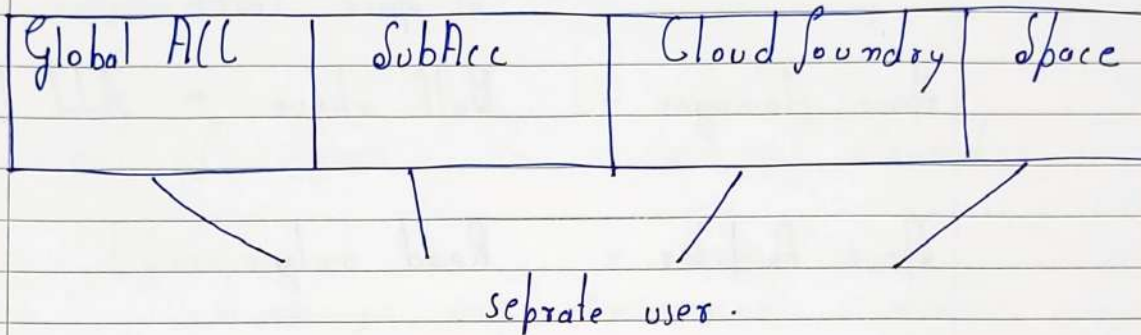
Dev Space > Instances > ——— bind to
Application

Access in Cloud foundry

Should be only possible when user
is available in org member under
cloud foundry section.

At Space

Space Level You have to create separate users.



Security

Org Member → Cloud foundry =>

Roles => Org Manager => Full Cloud foundry
Org Auditor => Cannot create Anything
only view

Org User => Don't work at cloud
foundry, user is a
space user that's why we
can see it.

Space Level

Space Developer - Deploy an application

Space Supporter - Managing Service, Binding at space level.

Space Manager - Both above + Add User

Space Auditor - Read only.

KyMa Environment  Kubernetes experience people should it.

Entitle the kyma in Subacc

Subacc Level Kyma Environment > Enable

Take - 20 min to Enable.

//_

Kyma is less abstract as compare to cloud foundry comparatively.

Kyma You can choose microservices You want to run although Cfoundry these services are compulsion

Cost of Kyma < cost of cloundry

Gardner

↓ manage API, Extension, module management.

You want to use btp Services so for that Gardner & BTP developed

BTP - operator > module.

#

Here - Namespace → space in Cfoundry

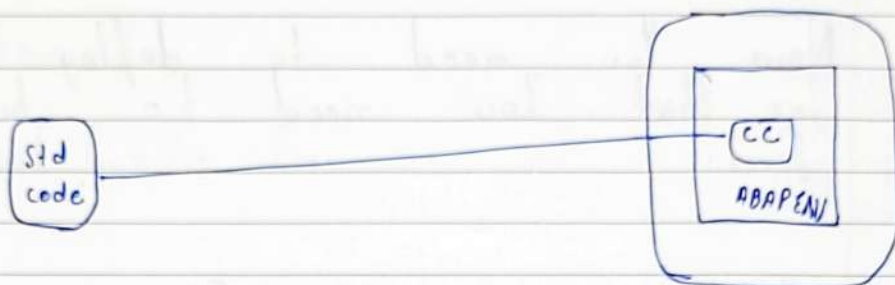
For deploying

You will need to upload a Yaml
file.

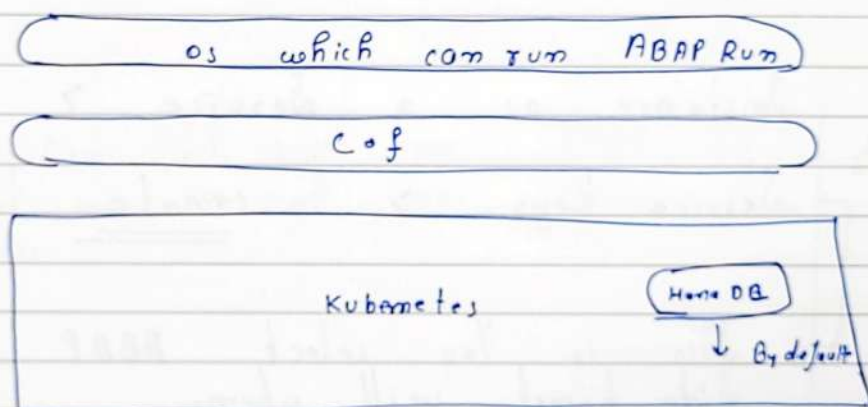
Then for URL API Interface.

ABAP Env

Extension development of SAP STD Service
Custom code migrate to cloud.



Technical Architecture



Eclipse.

When you create a key project it will ask you to create a project & where to run it.

> BTP ABAP Env

- Service key
 - Cloud foundry Env
- } choose

> Service key You can download & provide it.

> Cloud foundry provide API Endpoint or General overview

We can add user in ABAP Env for trial we cannot see. Cause its a shared access.

BTP CLI

By default our Windows command prompt does not know BTP CLI.

So we want to learn those

SAP Development Tools to download latest version of the btp CLI client for OS.

2> Extract using CLI -

windows	-	tar - xzf "_____"
Linux	-	tar - xzf "_____"
Mac	-	open a file

3> Copy client executable from unpacked folder to a directory of your choice

4> Set up Env variable for All for windows

5> open Terminal & Execute BTP Commands.

Command prompt:

> btp login

Syntax:

btp ACTION GROUP / OBJECT PARAMS

ACTION - list get create update delete
target assign move etc

GROUP - accounts security services

objects:

ACCOUNTS subaccount directory subscription resource-provider
entitlement available-environment available-region

SECURITY user role trust available-idp app role-collection

SERVICES offering instance plan platform binding

Command

★ btp target

Gives You Hierarchical Structure Use it to navigate

variants:

btp target	-- global-account	< subdomain >
"	-- subaccount	< subaccount Id >
"	-- directory	< directory Id >

★ btp get accounts / environment - instance envid

Get details.

btp get accounts / subacc → currently logged in subacc details

" subacc id → Get particular subid.

Next Section Skip →



Create

Subaccount `blp create account/subaccount --display-name "____" --region us10 --subdomain "____"`

Instance Service `blp create services/instance --offering-name alert-notification --plan-name standard --name alertnot`

Create Trust `blp create security/trust --idp <idpurl> --subaccount <subacc id> --origin <origin> --name <cust - IAS>`

Environment `blp create accounts/environment-instance --display-name <myenv> --environment cloudfoundry --service cloudfoundry --plan trial --parameters "{ \"instance-name\": \"you.org\" }"`

This name You can change.

Delete

Environnement

btb delete accounts / environment - instance
<environment id>

Dis

btb delete account/directory <Directory id>

User

http delete security / user <Email>

Assign

Role Collection

Role	idp assign	security / role - collection	"Global
Collection	Account Administrator	-- to - user	<example @mail>
	-- of - idp	<idp origin key>	

Entitled

blb assign accounts / entitlement -- global - account
 kb ----- > -- to - subaccount < ----->
 -- for-service xsuoa -- plan opiacess.

Important - When assigning memory in runtime.

MEMORY -- amount 1 → It will overwrite not add

Sub Acc - 50GB → 1GB
- 0GB → add 1GB

★ Update

```
blp update accounts/subaccount <subacc id>  
-- display-name " " --description "New-  
description" -- beta-enabled false
```

★ Move

Subacc to dir. blp move accounts/subaccount <sub id>
-- to-directory <dir-id>.

//_

↙ [Migration from SAML to Open ID]

★ Only through CLI

btb migrate security / trust <origin of SAML>

-- idp <TENANT>

backup is created. → ★

BTP Cloud Foundry CLI

> Setup

1. Go to CF CLI installation guide github repository
to install file
2. Extract or Install if Installer
3. Set the Environment variable for your account in windows machine
4. In Case of binary - Just extract and put it in a new folder & set some folder path in Environment variable for you windows account in your machine

> cf login

API End point You can
see in general
section
Subacc.

find directory

cf push <file name> -m 256M -k 256M

Space

cf create-space prod

Create Space Quotas we can do. —

Application Logging Services

Koi app run kar raha hai, Now
You need to monitor logs, Network,
users all that.

Subaccount > Create. Application Logging
Service.

Bind it to an app

You can see the detail Log's.

Audit Log Management System

Audit Log Services

Audit Data

Who is doing what why. All that
free for BCP Applications.

Some facts:

- ★ After 90 Days Audit Logs are deleted
- ★ Your company You need to take care it of
Your self.

- Audit Log retrieval Subacc - Default
 - " " Global acc - Central
 - Audit log retrieval & storage - premium
 - No 90 days limit
- PLAN ←

Category

audit • data • access } Data protection
audit • data • modification } & privacy

audit • security - events } Security Related
audit • configuration }

//_

> Create Instance for Audit Log Service

> Service Key Creation > in Instances & subscription.

Service bindings. — create

Copy Service

Concept

What You can see is.

Anything that is running on Cloud Foundry

Org Token is issued XSUAA server Issue a token.

O-Auth token is issued.

for fetching log You need Some When You Bound. Create You will see uaa

~~user=uaa~~, Authentication Server uol,
Certificate Keys are also there

//_

So if some 3rd party wants to access

- They will need OAuth token from uaa server
- Then they will access

Look into Service Key

uaa: {
 "url": _____

Step-1 Create Instance for Audit Log Service

Step-2 Create a Service Key

Step-3 Get Access token with post Call : <value of "uaa.url"> /oauth / token ? grant_type = client_credentials

Step-4 Get Audit Log using call get call :

<key-url> /auditlog / v2 / auditlog records

//_

In postman for Step 3

You will post request

url → url /oauth

You will provide Client id — user
Client secret — Password
under Auth. tab

You will Receive token.

Copy it → Get Request

Step 4 Key-url is Audit Log Management
Service URL.

Bearer token - under Auth

If You want params. You can add
time params also.

Time Parameter

time_from = 2023-01-09 T 10:42:00
time_to = 2023-12-10 T 10:46:00

Automate it

Integration Suite + Data Sphere

You can automate this.

B2P Integration

★ It help Application running on B2P to consume data, services or other resources from any on prem & cloud Target System

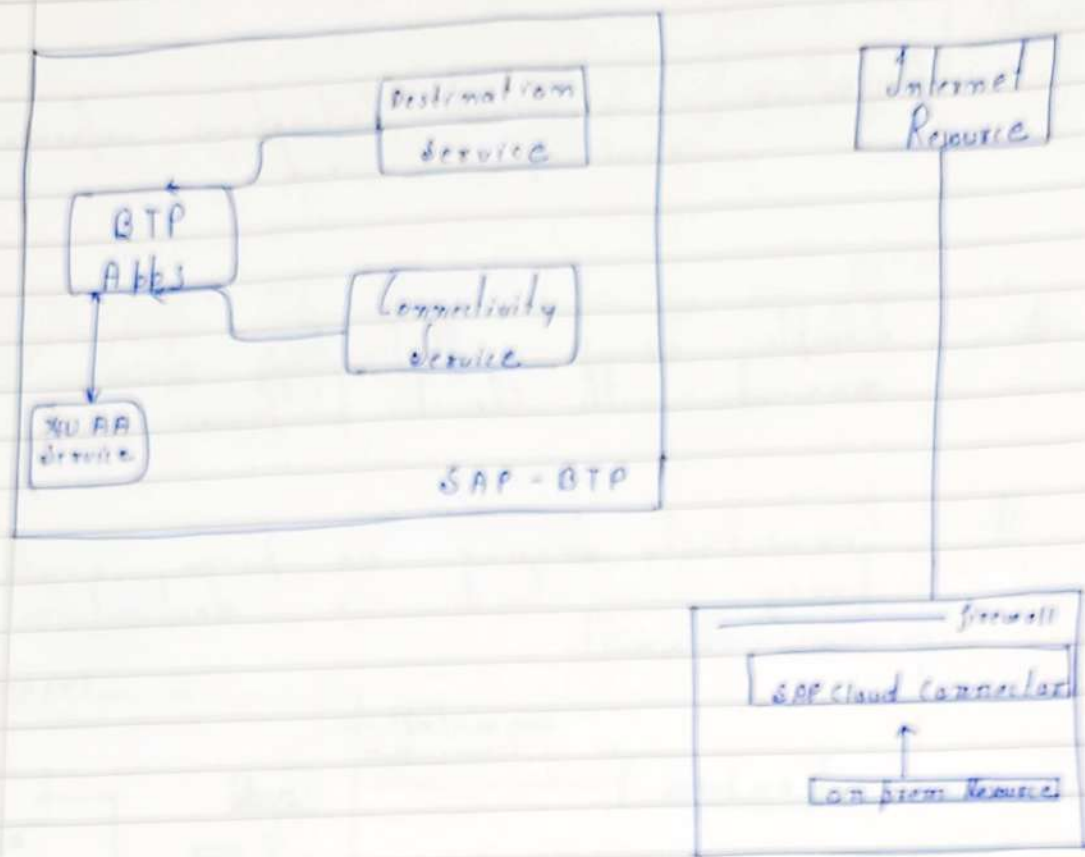
→ Connectivity Service
→ Destination Service } Microservices.

★ Connectivity Service + Destination Service both are required for on prem.

★ Destination Service is enough for cloud to cloud connectivity.

On Prem Connectivity - On prem Communication via HTTP or RFC for your cloud app

Destination - Contain technical info abt target system service or resources that are req to consume by b2b apps.



Cloud Connector

Cloud connector aapke Intranet mein
 hai bhi system mein ho sakte hain
 ki us aapke intranet mein to hain

fire wall ke andar.

The cloud connector runs as an on-premise agent in a secured network & act as a reverse invoke proxy between the on-premise network & SAP BTP

Cloud Connector is a tunnel if we have large amt of data we need to transfer b/w On prem & Cloud we could make sure to host it more storage.

★ In Connectivity section of Subacc we can see cloud connectors.

It shows how many cloud connectors are connected to it.

It doesnot mean cloud connectors are there in BTP — ✗

Integration

S-1

Cloud - Subacc

- Open cloud connector url in on prem
- Add Subaccount

↳ Details in subacc overview section
subaccount id
Email } user have role collection
Password }

We need cloud connector administrator role which is a part of following role collections -

- > cloud connector administrator
 - > connectivity & Dest admin
 - > Subacc admin
- } Any one

It must be S User Id.

★

[Password will be S User Id] -

Location Id -

Uniquely identify a cloud connection at Sub Acc level.

Save

Right now we have established cloud Connector on prem network to subacc. so any application on network is connected with subacc.

S-2

Onprem System — Cloud Connector

> Go to cloud connector open the connecti-

> 2 options:

Cloud to on-prem → Req is triggered by cloud.
Onprem to Cloud → Vice Versa

QTP want to fetch data - It will trigger
> cloud to on-prem.

→ Choose. → (7)

> Select System → S4 → ABAP System

> Choose protocol

> Internal host & port = ?? → SMICM

Log into target System, In SMICM.

Go to > Services

There are host details

> Virtual Host >

We provide real host & port prev
if we don't want that to appear
in BCP we can write anything.

that will show

> Next > Save

Click on Actions > ☒ → Reachable
should be there.

Under here as we add ABAP
System we can add Multiple Systems

- _/_/_
- ★ We have 2 options: HTTPS, Calls or RFC
 - ★ Webservices Calls — Odata, Idoc
 - ★ Function modules — RFC.

> For RFC — Till it Application Server & Instance no. will be available in SAP Logon. You can check

How can I access web service

web SIC F > Execute show us all on prem resources that we can target-

default host > we can see paths > sap > —

we can see what we can access click
we can also see a service.

You need to connect this on test in BCP

2

SE37 → RFC Junction module

RFC *

See what are diff RFC-func module we can access.

Now once this is established we cannot access all the Resources

BCP we can see Exposed Back-End System but no resource available

3

Resource Exposure

Cloud Connector > Bottom we can see in connection.

Resources. tab.

We need to add additional properties

10. client. ashost

" . client. client = ?? what client

" . client. sysnr = 'Instance no'

" . destination. pool_capacity =

" . " . pool_check_connection = 1
↳ check connection

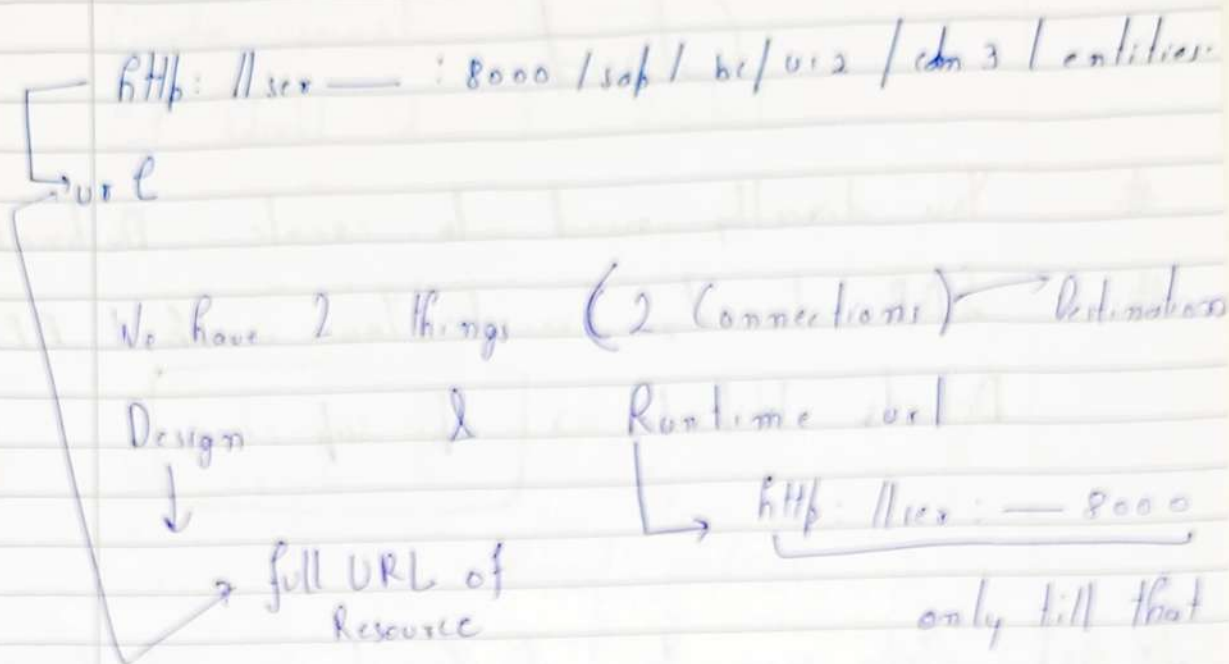
host id - we can see in cloud connector
copy url without port no.

5 th connection can be made

Save & Check.

Now we can consume this destination.

1. URL - url needs to be detailed



★ 2 Destinations need to be setup
refer to documentation for details.

Additional Properties — what are they
Check these

Design & Run Time Destinations

BCP Integration with Cloud

You directly need to create Destination

SAP already provide You with API

Details Website => api.sap.com

In use Make sure You specify Service protocol
—— .com/service/scim / Users —— all details on site

Proxy type - Internet

User & password

IAS ———> Its different

You — Add Administrator, Give Name
 system

Open System user

Create secret →

Get Client Id & Secret
↓
user & password

#

We create the destination with name as specified in documentation cause it is hard coded.

2/ Scenario



- BW2 wants to reach BP Automation.

- _/_/_
- BW2 needs to reach Automation Backend.

1 Automation Instance You Need

- Create a Service
- SAP Build Process Automation
- Standard plan
- Runtime - ~~other~~ — cloud foundry
- Instance Name - bpoapiaccess.

This is a Instance type of Subscription
You can create a Service Key Here

> —————> options create Service Key

2 Destination —————> Create in BW2 Subacc.

Name is specific => sap-process-automation-service.

type HTTP
URL RH
Ptype Internet
Auth 0 Auth 2 Client Credentials.

→ api Endpoint in BPA Credentials bottom.

uaa { client id —
client Secret, —
client password }
}

url — Token service url
} of that Server. to get token.

Help portal of BWZ

Additional properties. — mention

Any Application that is running on b/p we don't need destination & Runtime url or destination different.

BCP DEVOPS.

Develop $\rightarrow$ Version Control $\rightarrow$ Build, Test $\rightarrow$ Transport
 Section
 Acptl. Release

→ Deploy → Monitor

For Each Sub Acc You will Enable Cloud Foundry
to create space.

CCMS

- _/_/_
- # In CTMs we create node which are digital Replica of system Dev, Test, Prod
 - # Then accordingly You can connect System.

Setup

- Subscribe CTMS
 - Create destination for each subaccount which will work as node in TMS
- https://deploy-service.cf.us10-001.hana.ondemand.com → this will change
|sl|prot|<orgName>|<space-name>|slp
- It is cloud foundry api endpoint.

- Create Node
- Create Route.

Endpoint url to target space in cloud foundry

Prerequisite

For Node activation we need Destination to Dev, Prod — spaces.

First we need to create connection with spaces.

Step-1

Create destination to target Space

Name -
type - HTTP
Des -
URL - $https://deploy-service.<cloud\ foundry\ url\ from\ cf>/slprot/<org-name>/<space-name>/slp$

Basic Authentication

Some one with developer access in Space
→ Space Developer Role.

Create Node in CTMS

Transport Nodes > create.

Allow Upload > Means if You want to manually upload to a particular Node

Forward-Mode > on Success.

★ Virtual Node is when we do not reach out to space

★ Content type we can choose - Application -

★ Destination - Choose Destination

Create Routes - Transport Routes.

★ Now in Dev mode Integration Suite will push code to CTNS Dev Node. No destination is req for Dev Node.

In Integration we will say push code to CTNS. Destination config will be different

Name	-	Transport Management Service	} same process create instance, service key.
type	-	HTTP	
URL	-	_____	
Proxy	-	Internet	
Authentication	-	OAuth2 Credentials	
Token Service URL	-	_____	Dedicated =
URL	-	_____	=> add /oauth/tokens

AP

sourceSystemId - Dev-node } Node name.
node-name = Dev-node }

Security Management

* In a Gsp Management thing. -:

→ There is a convention that AppName - Profile of User.

Ex BAS - Admin - User.

Import Users

* To Check Users =>

for file format You can Go to users & Authorization > Export => there will be a csv format. Remove column You don't want.

Sendmail Column for sending mail TRUE / FALSE

Application

How →

//_

1 Application - choose Your Application -

> Authentication & Access

> User Application Access as

Public
Internal
Private

2 Public means he can Register what are details
Some User Attribute for Access.

3* Risk Based Authentication - we can create
rules on various
things.

TOTP - Scan just like Microsoft Authenticator

↳ SAP Authenticator App - 2 factor Authentication

Web Authentication - Passkey store in device.
Device will Register.

4* There Can be conflicting user rules so we
need to prioritize.

SAP CIS As Proxy.

MS Azure AD — SAP IAS — BTP App

Microsoft Entra Id → Trust Setup

- * Login to MS Entra Id
- * Enterprise Application → Create → SAP
→ SAP CIS → Name Create
- * Same we need to do in CIS.
- * Identity Providers > Create > Name, SAML 2.0,
↓
AD support
- * Go to placeholder in AD there select
Single Signon > SAML
- * Download Metadata of IAS; Tenant Settings
- * In AD upload
- * for AD Metadata Federation XML In ③
- * In IAS placeholder under Identity provider
SAML 2.0 Config.
- * Choose Identity provider type Microsoft ADFS.

→ our placeholder / /

* Go to Azure IAD → (IS Enterprise App → properties

Assignment Required - No

* Allow it for a particular Application

Application > SubAccount > Conditional Authentication > Default Identity provider — change.

* Shadow user will be created in BCP.

* Attributes to send we can control.

→ SAML - Single Sign On - ② Attributes & Claim

→ Add New claim, Add Group Claim

→ We can get Group-Id from ID & Can map to role collection.

IPS

Read User
from System



Write User
from System

> Create Automatic User in Target Application

> Filter User & Grp we can transfer

> Src-System Read user — Write user in Target Syste

• It can also act as a proxy with IAG.

Security Artifacts

User → Rolecollection → Roles ^{based on}

Scope ^{many} ← Role Template → Attribute

↓
what can he do

↓
Rules like if he
is a HR dep.
(User Attrib)

Principle Propagation

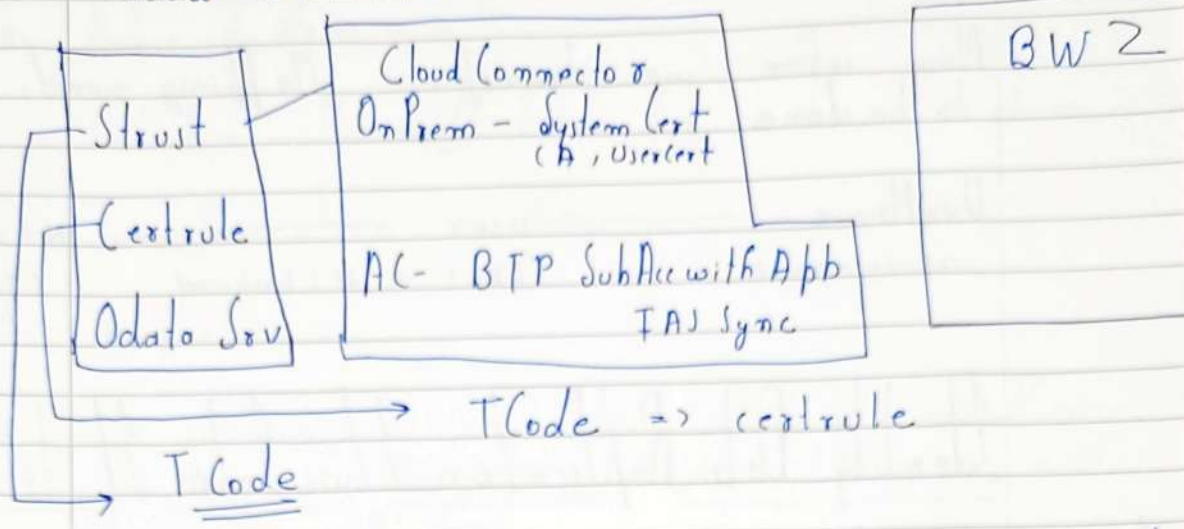


Through App1 You access App2 page Similar to overlay or iframe.

App1 ke through App2 ko page access karo toh aapki identity App2 pe jaati hai to check but dono mein Pkey honi chahiye to uniquely identify.

*

Cloud Connector → Trust BTP IAS



Upload a certificate created on cloud connector to on prem.

Certificate => Connector → Configuration → create

User Certificate => Also Required

CA Certificate => Also Required.

The user mail is configured in on prem SSO!

Also

IPS

Now when we transfer Mapping need,
to be done

User Name ————— user ————— Name
column name (IDS) → IPS backend (Target backend)

Identity Data Replication b/w Entra Id to IAS

1

Users in Entra Id :- Department, Employee Id, City,
Country, Email, Mobile phone.

2

Create IPS placeholder ⇒ use Graph API

- * AD → AppRegistration → New Registration (Create a Tech User)
- * API Permissions > Microsoft Graph > Application Permissions >
- * Select user - Read Write All, Group Read Write All, Dir Read Write All
- * Grant Admin consent for default directory.
- * Copy Application Id, Create a secret - Copy
↓
copy value
- * Goto Main page copy primary domain.

3 IPS Connection

* Source System - :

Details - Microsoft Intra Id

Save

Transformation is there.

* properties - :

See who can see in here attributes are there
user, group that IPS Export if they are not
there job will fail.

* properties →

Type	-	HTTP
ProxyType	-	Internet
URL	-	https://graph.microsoft.com
Authentication	-	BasicAuthentication
User	-	Client Id / App Id
<u>credential</u>	← Password	- Secret

aad.domain.name - primary domain
OAuth2TokenServiceUrl : http://login.microsoft.com /
<primary-domain> /oauth2/tok

ips.trace.failed.entity.content : true

ips.trace.skipped.entity : true

ips.trace.skipped.entity.content : true

4

Some You Need to do for IAS

Create a technical user , provide Authorization
Create secret & password.

→ IPS Create Target System (Choose Source

→ Type - HTTP

ProxyType - Internet

URL - < IAS URL >

Authentication - Basic Authentication

User - < IAS System User >

credential ← Password - < IAS System User Password >

5

Src System - Job - Run.

6

If You want to read Groups → Src System
→ Transformations > group - ignore: False

7

Check Provisioning logs - to check job.

8

Make Sure all attributes that You are reading
have their transformation

SAP IPS HELP PORTAL